

Algorithms and Complexity

Module 1

Automata Theory

Lesson 1.0 Introduction

- Course structure and navigation
Reading • 15 min
- How to learn effectively on this course
Reading • 15 min
- Introduction to the course
Video • 55 sec
- Course Syllabus
Reading • 10 min

Lesson 1.1 Finite automata

- Introduction
Video • 1 min
- Basic definitions, letters, strings
Video • 4 min
- What is an automaton?
Video • 8 min
- Basics
Practice Assignment • 30 min

Lesson 1.2 Deterministic automata

- Finite automata – example (part 1)
Video • 3 min
- Finite automata – example (part 2)
Video • 8 min
- Working with Automata Simulator
Video • 6 min
- Design an automata to accept a simple language using Automata Simulator
Dialogue • 30 min
- Language of the automata
Video • 4 min
- Recognise a language
Video • 4 min
- Exercises with hints and tips
Reading • 1h

Lesson 1.3 Non-deterministic Automata

- Deterministic finite automata (DFA) vs nondeterministic finite automata (NFA)
Video • 4 min
- DFA example
Video • 3 min
- DFA example in Automata Simulator
Video • 4 min
- DFA Simulator
Lab • 20 min
- Computation by NFA
Video • 3 min
- NFA example
Video • 5 min
- NFA example in Automata Simulator
Video • 2 min
- NFA Simulator
Lab • 20 min
- Exercises with hints and tips
Reading • 10 min

Lesson 1.4 Summary and Assessments

- Conclusion
Video • 3 min
- Check your understanding: End of module 1
Graded Assignment • Grade: 100%

Module 2

Searching and Sorting Algorithms

Lesson 2.1 Introducing Algorithms

- What is an algorithm ?
Video • 4 min
- Tell us about an algorithm that you use
Discussion Prompt • 20 min
- Representing algorithms
Video • 8 min
- Simple algorithms – insertion sort
Video • 5 min
- How else would you sort?
Practice Assignment • 30 min
- Simple algorithms – bubble sort
Video • 5 min

Lesson 2.2 Sorting and Search Techniques

- Selection sort
Video • 4 min
- Insertion, bubble and selection sort
Practice Assignment • 25 min
- When we should use which sorting algorithm?
Dialogue • 15 min
- Sorting algorithms simulator: Bubble Sort and Insertion Sort
Lab • 20 min
- Exercises with hints and tips
Reading • 10 min
- Search techniques
Video • 4 min
- Binary and sequential search
Practice Assignment • 12 min

Lesson 2.3 Heap sort

- Binary trees and heaps
Video • 3 min
- Heapify algorithm
Video • 4 min
- Heap sort
Video • 6 min
- Binary trees and heaps
Reading • 1h
- Heap sort
Practice Assignment • 20 min

Lesson 2.4: Summary and Assessments

- Conclusion
Video • 1 min
- Check your understanding: End of module 2
Graded Assignment • Grade: 100%

Module 3

Recursive and Iterative Algorithms

Lesson 3.1 Recursive and Iterative Algorithms

- Recursion
Video • 7 min
- Iterative algorithms
Video • 4 min
- Recursive algorithms example
Video • 5 min
- Recursive and iterative algorithms
Practice Assignment • 25 min
- Quick sort
Video • 6 min
- Quick sort example
Video • 4 min
- Quick sort
Practice Assignment • 20 min
- Partition and quick sort algorithms
Reading • 1h 25m
- Lesson 1 exercises with hints and tips
Reading • 10 min

Lesson 3.2 Merging

- Merging lists
Video • 4 min
- How does this magic work?
Practice Assignment • 30 min
- Merge sort
Video • 4 min

Lesson 1 exercises with hints and tips



Now it's time to put into practice the concepts that you have learnt so far in this lesson. Attempt the following exercises. If you get stuck, you can refer to the hints and tips at the end of the page.

Please note that these exercises are optional, further practice – but we strongly recommend that you engage with them in order to test your knowledge and see where you might need to do additional study.

- Consider the following list:
A: 13, 1, 16, 8, 2, 9, 4, 5
Using quick sort, show A after one iteration (in ascending order). Show A after the second parsing/iteration.
- Consider the following list:
A: 13, 1, 16, 8, 2, 9, 4, 5
Show the list after calling partition.
- Do you think it is important if you consider the first element as the pivot in the partition algorithm? Why?
- Write a recursive algorithm to reverse a string recursively.

Hints and tips for each question:

- First, run the partition by selecting the first element as the pivot. Then, call quick sort twice. This is the first iteration. For each left and right sub-lists, call the partition; this would be the second iteration.
- Consider 13 as the pivot and trace the partition.
- No. Try to find out why.
- Check the size of the string,. If it is 1, just return the string, otherwise return reverse (str[2..n])+str[1]. Str is the given string. String[1] represents the first letter and str[2..n] represents the remaining string after deleting the first letter.

Go to next item ✓ Completed

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